# Intake Test

Does the intake pull balls in, does the water wheel lift them, and do the two switches spot a NECTAR? (issue #4, decisions 024, 025, 028)

| | | | |
|---|---|---|---|
| Names | | Date | |
| Battery voltage at start | | Battery at end | |

**OpMode:** Intake Test (Tests group on the Driver Station). **Needs in the config:** `intakeServo`, `waterWheelServo` (Continuous Rotation Servo), `nectarDetectorLeftSwitch`, `nectarDetectorRightSwitch` (Digital Device)

**Safety:** robot on blocks or on the floor with space around it, hands clear of moving parts, one person ready to press STOP. Nothing moves during INIT; everything starts after START.

## Controls (gamepad 1)

| Button | Does |
|---|---|
| A / B | spin ON-OFF (toggle) / STOP. Starts stopped |
| DPAD up / down | speed of BOTH +/- 0.05. The speed is not saved: copy the final number into Intake.java |
| DPAD left / right | direction (toggle): IN + water wheel UP, or OUT + water wheel DOWN |
| X | which one runs: BOTH, INTAKE only, WATER WHEEL only. Use "only" to check one at a time |
| (no button) | the switches are read all the time, even in INIT. Watch the telemetry lines "NECTAR detected", "Nectar left switch", "Nectar right switch" |

## Steps, in order

| # | Do this | Expected | Actual (what happened) | If wrong |
|---|---|---|---|---|
| 1 | X to INTAKE only, press A (direction IN) | balls go IN | | `INTAKE_REVERSED = true` |
| 2 | X to WATER WHEEL only, press A (direction UP) | balls go UP toward the launcher | | `WATER_WHEEL_REVERSED = true` |
| 3 | X to BOTH, A on, 3 balls. Tune the speed with dpad up / down | no jam, no ball flying out | | copy the speed from the screen into `INTAKE_SPEED` / `WATER_WHEEL_SPEED` (0 to 1) |
| 4 | Nothing touching the levers | both "not pressed", "NECTAR detected: no" | | says PRESSED: switch `NECTAR_SWITCH_WIRING` to the other value |
| 5 | Press the LEFT lever by hand | "Nectar left switch: PRESSED", "NECTAR detected: YES" | | never changes: try the other channel of that digital port (n / n+1) |

| # | Do this | Expected | Actual (what happened) | If wrong |
|---|---|---|---|---|
| 6 | Press the RIGHT lever by hand | same, for right | | same |
| 7 | Run a NECTAR (big ball, ~3.6 in) through | YES while it passes | | levers too far apart: move them in |
| 8 | Run a POLLEN (small ball, ~2.8 in) through | stays "no" | | levers too close: move them out |

Step 7 and 8, repeat 5 times each. NECTAR detected: ___ / 5. POLLEN wrongly detected: ___ / 5.

## Numbers to put into the code: `subsystems/Intake.java`

| Constant | Was | New value | Checked by |
|---|---|---|---|
| `INTAKE_REVERSED` | true | | |
| `WATER_WHEEL_REVERSED` | true | | |
| `INTAKE_SPEED` | 1.0 | | |
| `WATER_WHEEL_SPEED` | 1.0 | | |
| `NECTAR_SWITCH_WIRING` | NORMALLY_CLOSED | | |

## Wiring, for `hardware-config.md` (Intake table)

| Device | Hub (CH / EH) | Port |
|---|---|---|
| `intakeServo` | | servo |
| `waterWheelServo` | | servo |
| `nectarDetectorLeftSwitch` | | digital |
| `nectarDetectorRightSwitch` | | digital |

**After the test:** a coder puts the new numbers into the constants and removes the matching TODO comments; the results go on GitHub issue #4. The filled-in sheet goes with the portfolio entry `docs/portfolio/2026-11-04-intake-and-nectar-detection.md`. Photos of anything surprising!